InHand Networks

Intelligent Management. Seamless Connectivity.



InCloud Manager

- AI Network Assistant
- Zero-Touch Cloud Management
- Cloud-Native Management
- Remote Deployment

1. Product Overview

Intelligent Management. Seamless Connectivity.

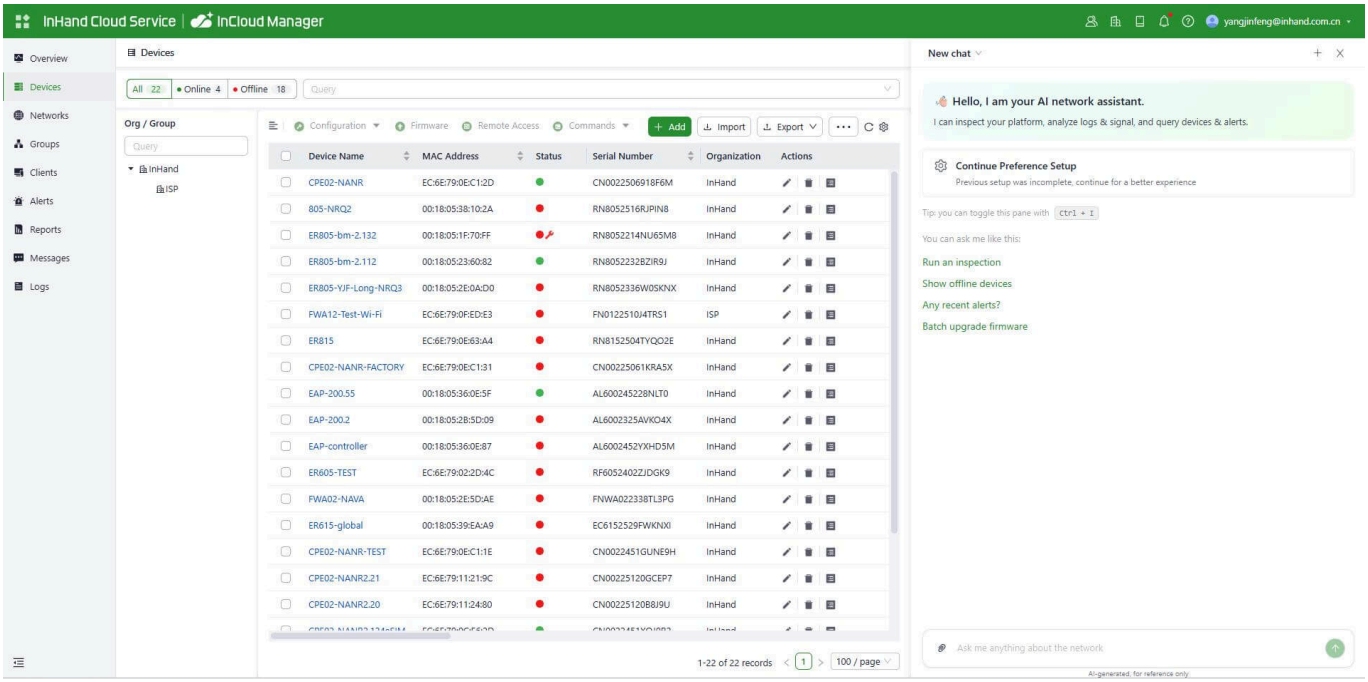
AI Network Assistant · Zero-Touch Cloud Management

InHand Networks InCloud Manager is an enterprise network management platform designed for retail stores and SMB offices. Combined with InHand edge routers, APs, and switches, it delivers a simplified, cloud-native network management solution. InCloud Manager enables businesses to move device and network operations to the cloud, supporting one-stop remote deployment and centralized management of network and communication devices across any scenario.

The newly launched **AI Network Assistant** brings network operations into the natural language era — issue plain-language instructions to complete device inspections, fault diagnosis, and batch configurations, dramatically improving operational efficiency. It helps businesses grow effectively, reduce management costs, and strengthen their competitive advantage in digital operations.

2. AI Network Assistant

Simply type a natural language instruction in the AI Network Assistant dialog to manage the full device lifecycle — no page-by-page navigation required, no professional network engineering background needed.



3. The O&M Challenge: Staffing Constraints and Efficiency Bottlenecks

Common pain points in traditional network management:

Channel Partners	End Customers
One person can only manage a limited number of devices — scaling up means falling behind	Routine inspection of dozens of devices requires hours of clicking through Web UIs
Experienced network engineers are scarce and costly	Troubleshooting an offline device means jumping through multiple pages — slow and tedious
Customers are geographically dispersed, making on-site visits expensive	Signal quality requires manual lookup and comparison against standards
	Batch operations are done entirely by hand, device by device

4. 9 AI Capabilities Covering the Full Device Lifecycle

Device Monitoring

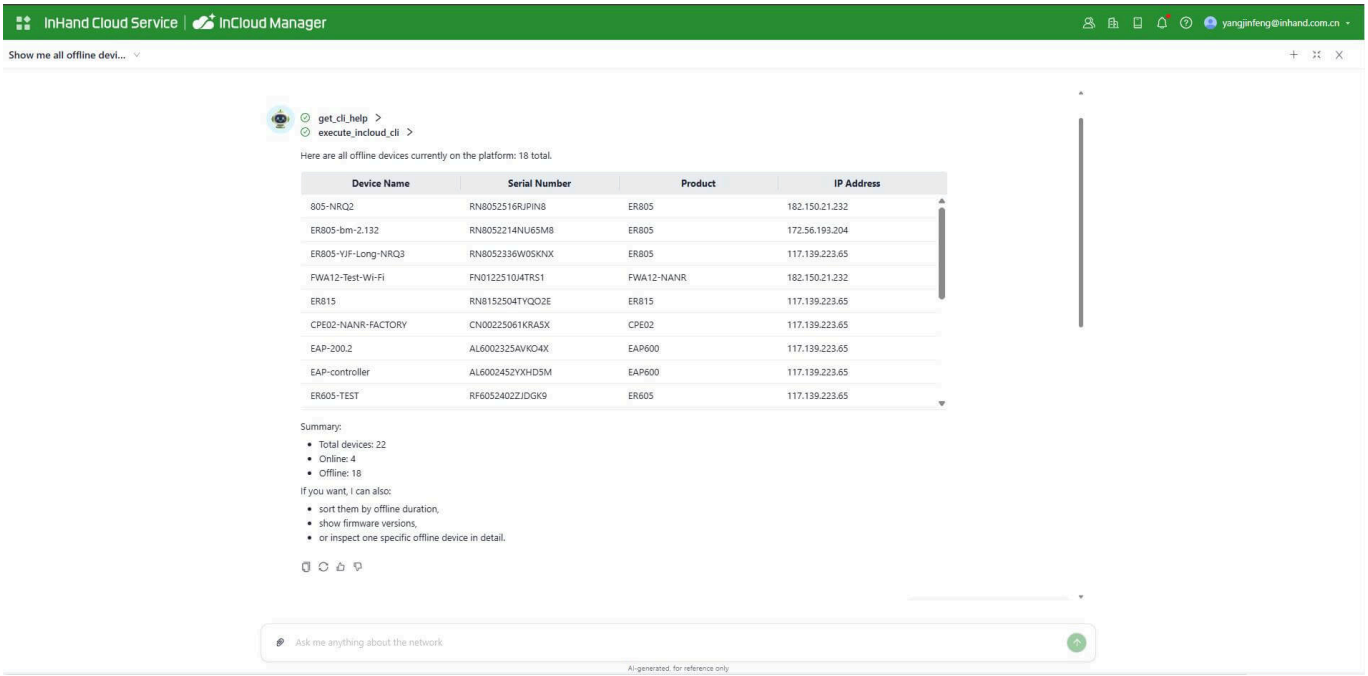
Get real-time visibility into every device's operational status — no manual page browsing required.

- Query online / offline status with filtering by name, location, or group
- View online history to track recent device stability trends
- Real-time traffic statistics to detect abnormal bandwidth consumption
- View connected client count and details

Example:

> User: Show me all offline devices
>
> Assistant:

Device Name	SN	Offline Since
Router A	SN001	5 minutes ago
Router B	SN002	5 minutes ago



Fault Diagnosis

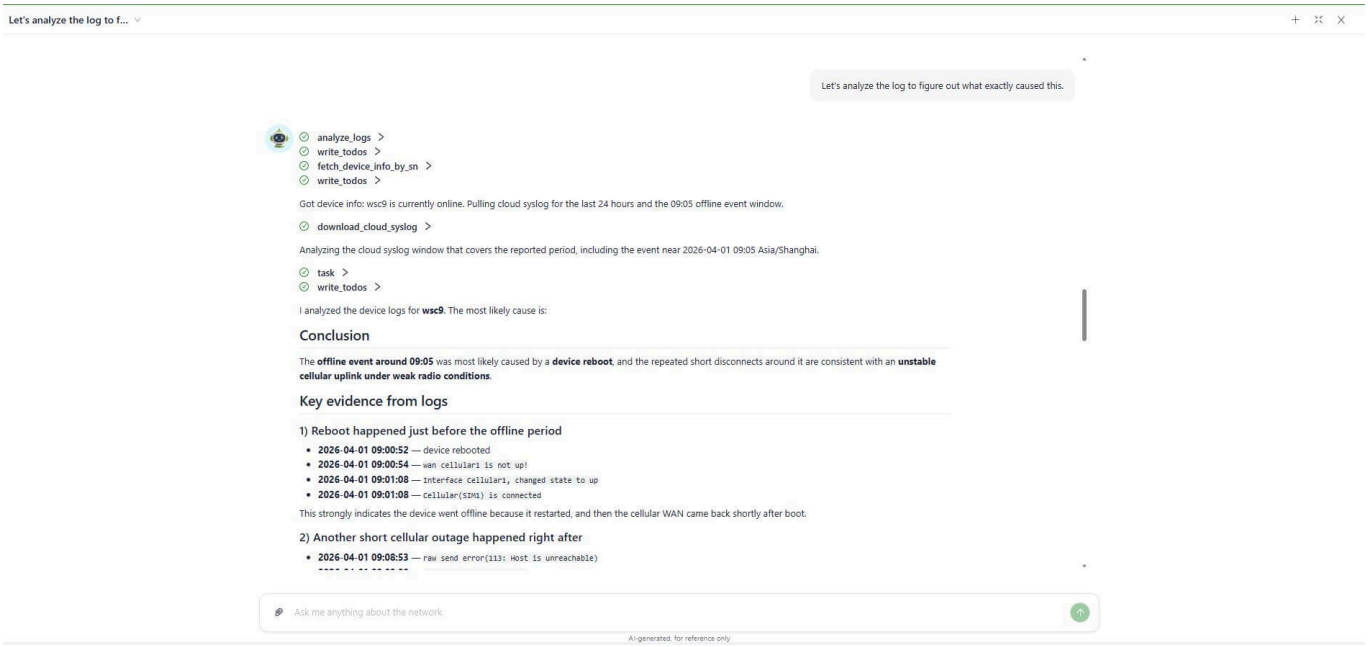
More than a simple query — the AI automatically runs a multi-step diagnostic chain and delivers a complete analysis with actionable recommendations.

Example: **Device Offline Diagnosis** — AI executes 5 automated steps:

Step	AI Action	Finding
1. Check device status	Confirm offline timestamp	Last online at 10:23
2. Review online history	Analyze recent disconnect patterns	3 dropouts in the past week, mostly at night
3. Check signal records	Analyze RSRP trend	Dropped from −95 to −112 dBm (weak signal range)
4. Review system logs	Search for key events	Modem reset events found
5. Conclusion	—	**"Persistent signal degradation causing frequent disconnects — recommend inspecting antenna"**

Example:

> User: Why did the first device go offline?
>
> Assistant: Auto-checked signal → reviewed logs → Conclusion: Signal degradation (RSRP −112 dBm). Recommend checking the antenna.



Signal Analysis

No need to manually look up standards or compare raw values — the AI evaluates cellular signal quality and delivers a clear, direct verdict.

Signal Level	RSRP Range	Description
Excellent	> −85 dBm	Strong signal, stable operations
Good	−85 ~ −100 dBm	Normal operation
Fair	−100 ~ −110 dBm	Potential risk, recommend monitoring
Poor	< −110 dBm	High risk, immediate action recommended

- Automatically flags devices that fall below professional thresholds
- View historical signal trends to detect degradation before it causes issues
- Cross-device signal quality comparison to quickly identify underperforming sites

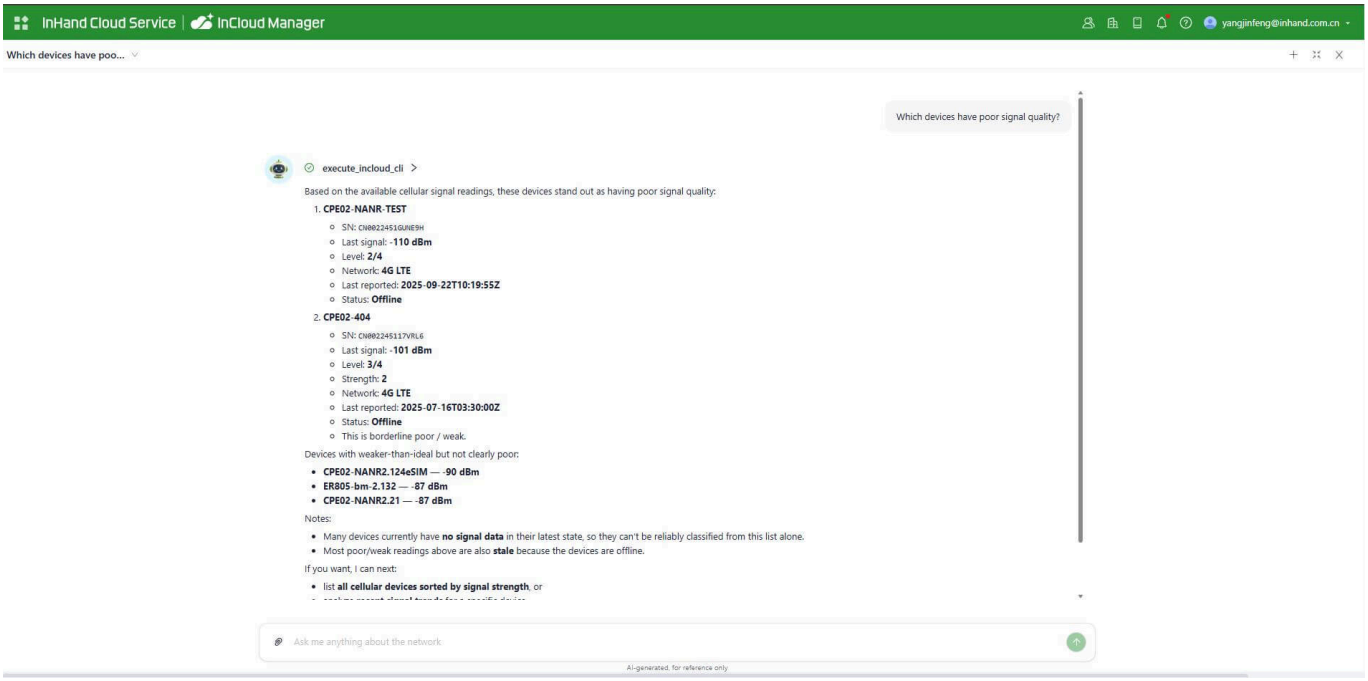
Example:

User: Which devices have poor signal quality?

Assistant: 3 devices currently have poor signal quality:

Device	RSRP	Level	Recommendation
Store A – Router	−118 dBm	Poor	Check antenna or adjust mounting position
Store C – Router	−113 dBm	Poor	Check antenna

Device	RSRP	Level	Recommendation
Warehouse B – Router	–106 dBm	Fair	Continue monitoring



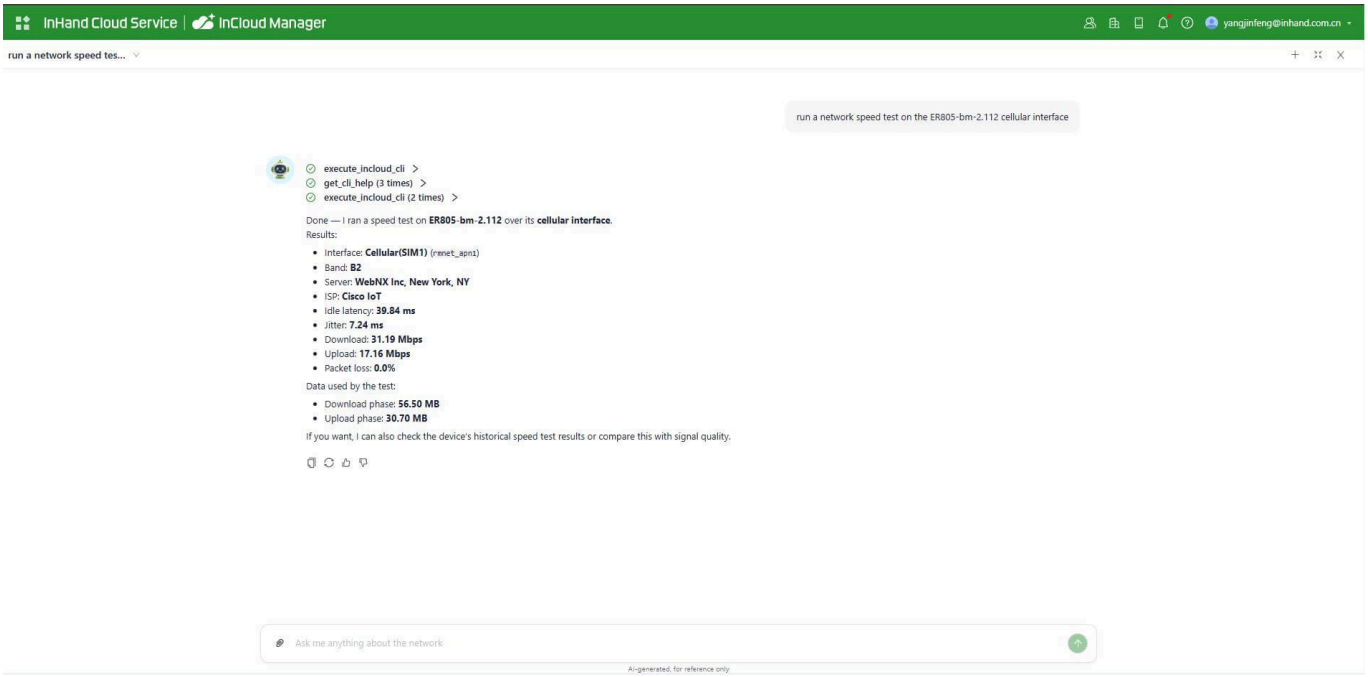
Remote Diagnostics

Run diagnostic commands directly without logging into the device Web UI — results returned instantly.

- **Ping**: Test network connectivity and latency
- **Traceroute**: Trace path and locate breakpoints
- **Packet Capture**: Capture packets for protocol analysis
- **Speed Test**: Measure actual bandwidth and verify link quality

Example:

> User: Run a speed test on Store A's router cellular interface
>
> Assistant: Speed test complete. Results:
>
> - Download: 48.3 Mbps
> - Upload: 12.1 Mbps
> - Latency: 34 ms / Packet Loss: 0%
> - Conclusion: Network is operating normally — current link meets business requirements



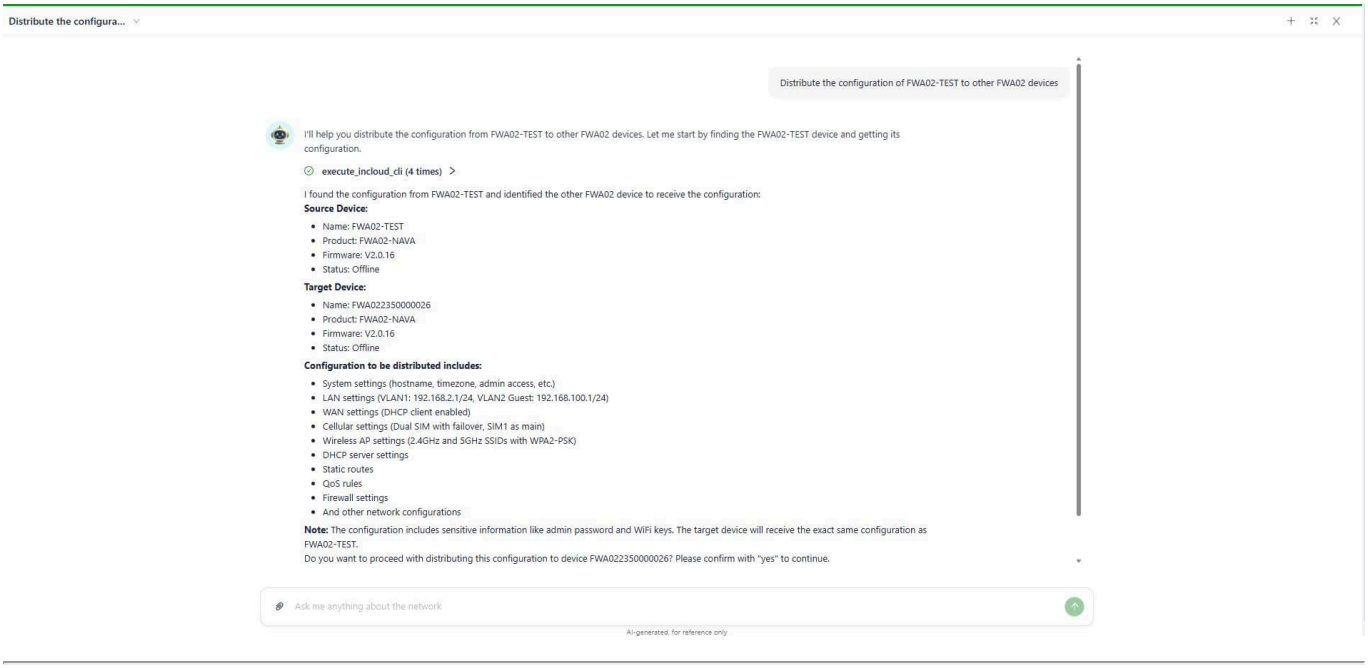
Configuration Management

Read, modify, and copy device configurations with full traceability and rollback support at every step.

- **Read / Modify**: Query or update device configuration items directly, without opening the Web UI
- **Copy Config**: Quickly replicate one device's configuration to other devices
- **Snapshot Backup**: Automatically save a configuration snapshot before any change
- **Rollback**: Restore to any previous configuration version in one click

Example:

> User: Use Device-xxx's config as the standard template and push it to all other ER805 devices
>
> Assistant: Current config comparison → Diff review → Confirm changes → **Awaiting confirmation before execution**



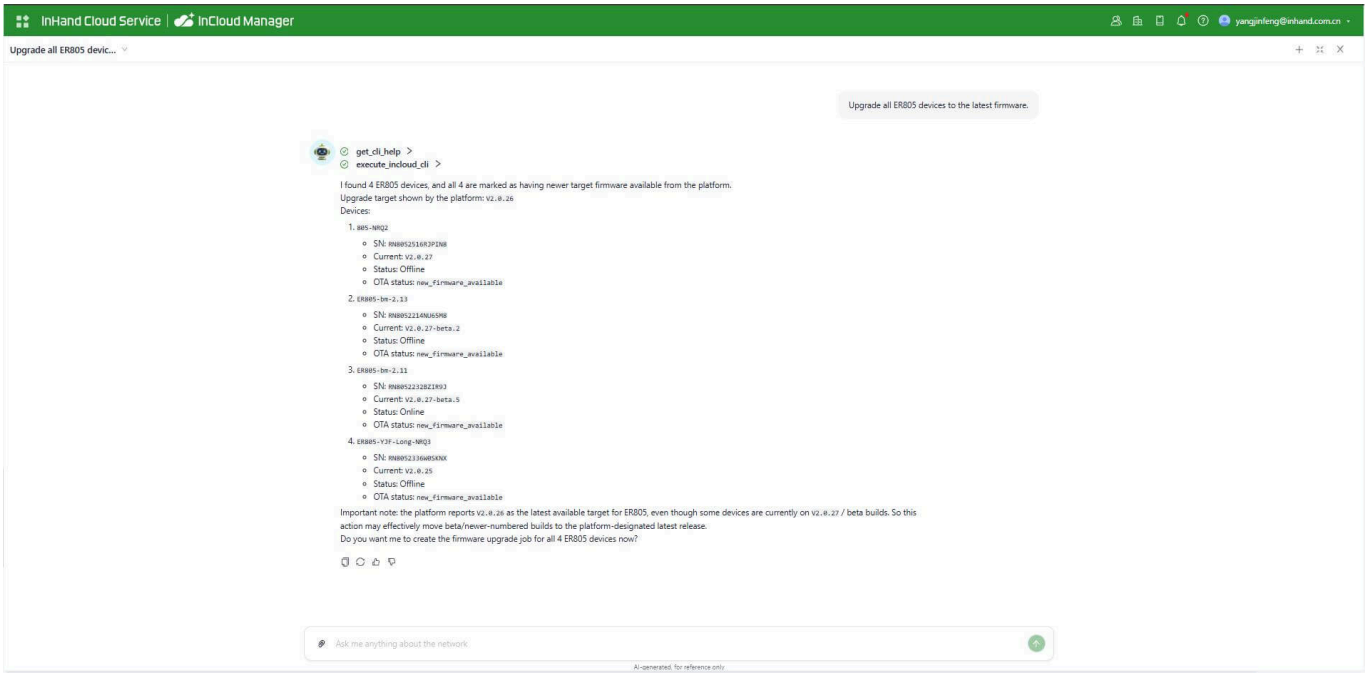
Firmware Upgrade

AI manages the entire process from version selection to upgrade completion, significantly reducing the risk of manual errors.

- **Compatibility Check**: Automatically verify firmware compatibility with device model before upgrading
- **Staged Rollout**: Support group-based or time-window-based batch upgrades to minimize business risk
- **Progress Monitoring**: Track upgrade status for every device in real time with automatic alerts on failure
- **Rollback Protection**: Instantly revert to the previous firmware version if an upgrade fails

Example:

> User: Upgrade all ER805 devices to the latest firmware
>
> Assistant: Detected 23 ER805 devices. Current version: V2.0.18. Latest version: V2.0.27. Compatibility check: All passed. Recommended: 3 batches of ~8 devices each, with 10-minute intervals. Estimated total time: 35 minutes. Confirm to proceed?



Alert Management

Centralized view and management of all network alerts — no anomaly goes unnoticed.

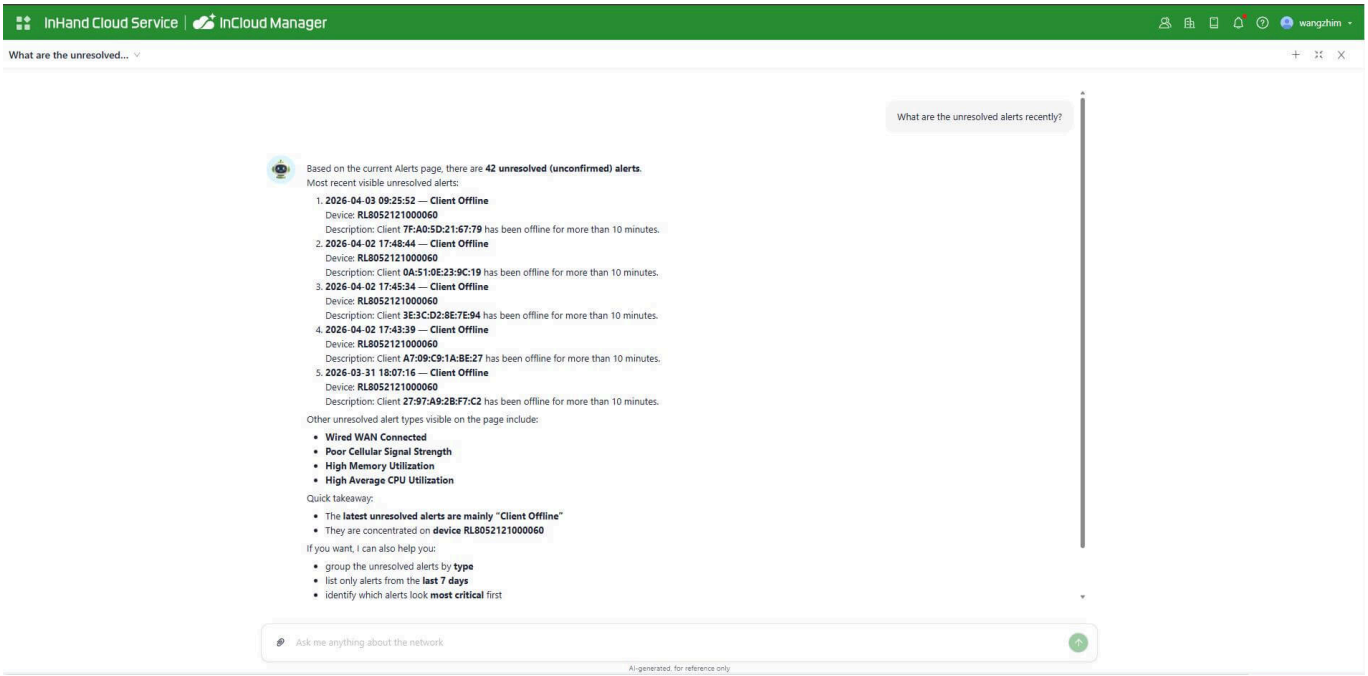
- View current unacknowledged alert list, sorted by severity
- Acknowledge / close alerts with action logging
- Manage alert rules: add, modify, enable or disable alert policies
- Correlated analysis of alerts and device status

Example:

User: What unacknowledged alerts are there recently?

Assistant: 5 unacknowledged alerts currently:

Device	Alert Type	Time	Severity
Store G – Router	Device Offline	Today 09:15	Critical
Warehouse B – Router	Cellular Data Overrun	Yesterday 23:40	Warning
Store D – Router	High CPU Utilization	Yesterday 18:22	Warning
Store B – Router	Link Status Abnormal	Yesterday 14:05	Info
Store F – Router	Signal Quality Degraded	Yesterday 11:30	Info
Would you like to handle them one by one, or prioritize a specific device?			



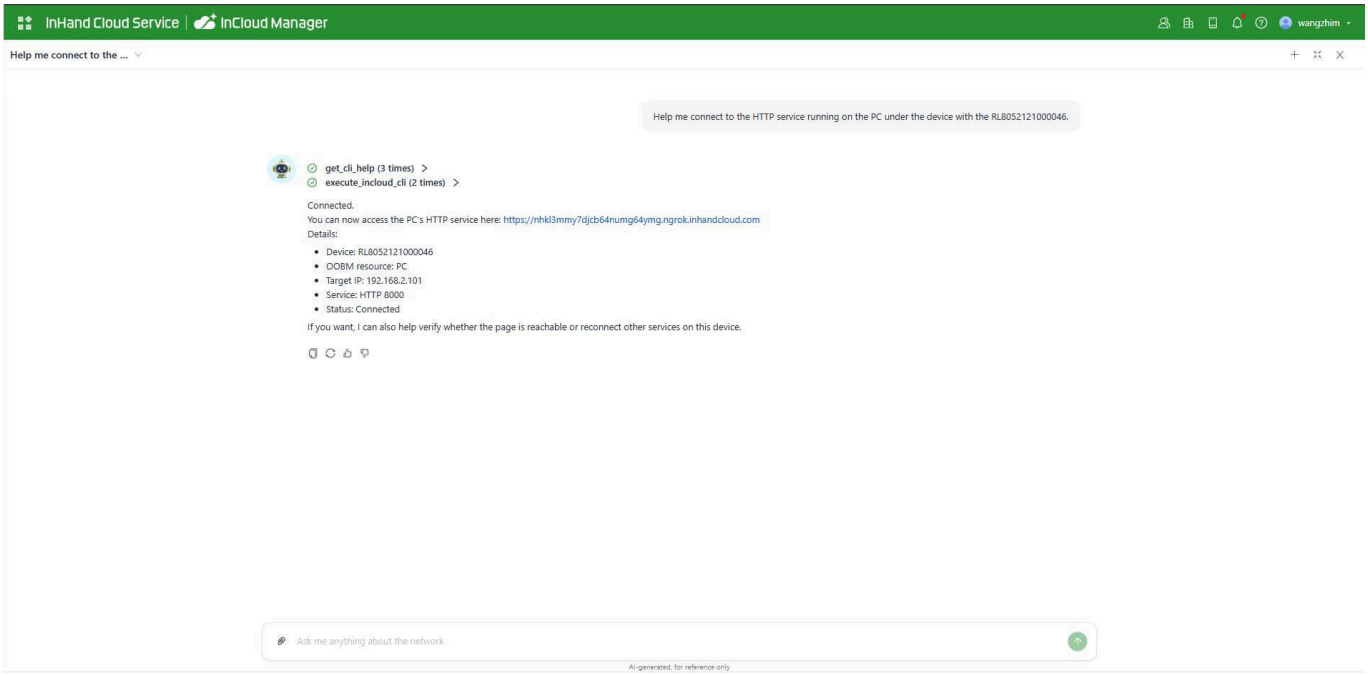
Remote Terminal

For advanced operations, launch the device control interface directly — no manual navigation required.

- **Open Device Web UI**: Jump directly to the target device's management page
- **Open Command Line**: Access the device CLI directly for advanced operations
- All operations are authenticated by user account; activity logs are fully auditable

Example:

> User: Open the management interface for Store A's router
>
> Assistant: Remote connection established. Store A (SN: ER805-001) Web UI is now open in a new window. This session has been recorded in the activity log.



Batch Operations

Issue instructions to multiple devices simultaneously — no repetitive manual device-by-device execution.

- **Batch Inspection**: Run a health check on all devices at once and generate a consolidated report
- **Batch Config Push**: Push a configuration template to a specified device group
- **Batch Firmware Upgrade**: Initiate bulk firmware upgrades with unified progress tracking
- **Batch License Management**: View license status, renew, or assign licenses in bulk

Example:

User: Run a monthly inspection on all retail store devices

Assistant: Inspecting 60 devices... Complete. Report:

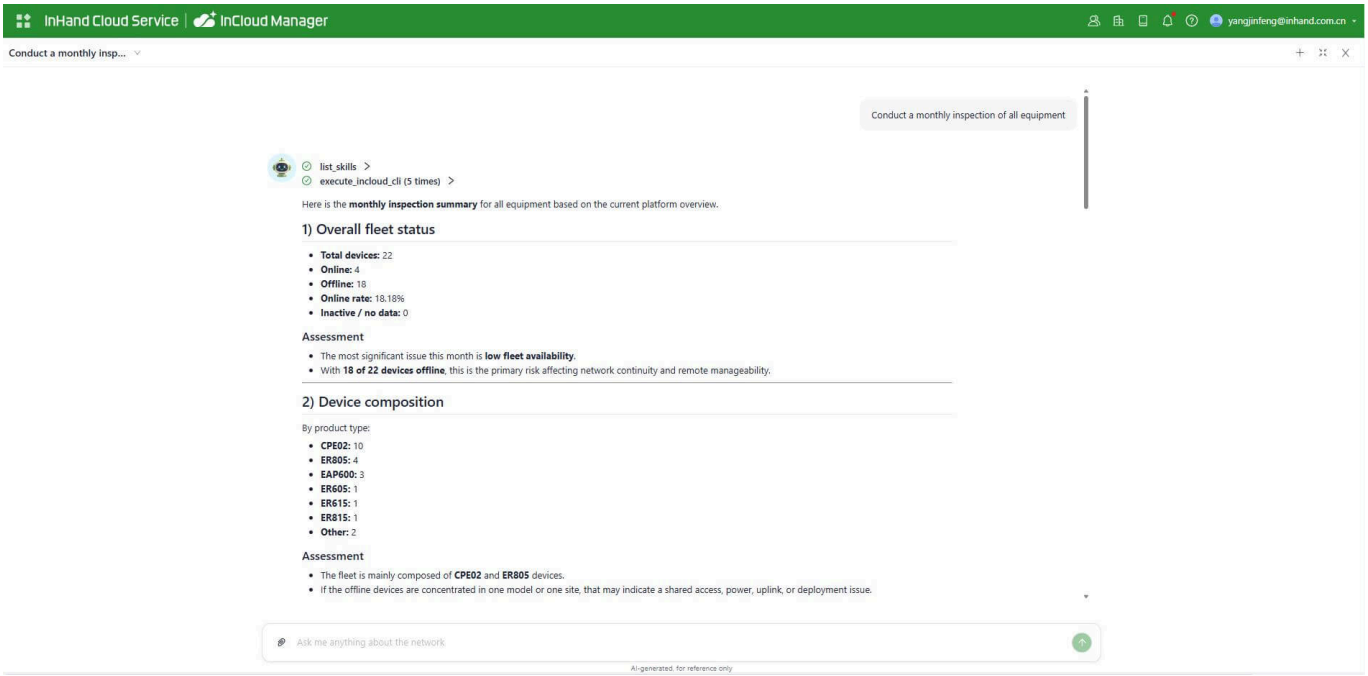
- Online and healthy: 42 devices
- Weak signal (attention needed): 3 devices (Store A, C, F)
- Offline: 18 device

...

- I can continue with one of these:

1. **Inspect the 18 offline devices**
2. **Check which device has config error / pending config**
3. **Review firmware candidates for upgrade**
4. **Focus on a specific device from the list**

If you want, I can start by drilling into the **offline devices** first.



5. AI-Powered O&M: Across-the-Board Efficiency Gains

Task	Traditional Approach	With AI Network Assistant
Monthly device inspection	Log into each device's Web UI individually — takes days	Natural language instruction — complete report in minutes
Offline device troubleshooting	Browse multiple pages manually, make your own judgment	AI auto-runs full diagnostic flow, delivers complete conclusion
Batch config deployment	Manual filtering, device-by-device execution	Issue one instruction, batch execution runs automatically
Bulk firmware upgrade	Manual device-by-device operation, manual progress tracking	Unified initiation, staged execution, automatic progress reporting
Signal quality audit	Manually look up data and compare against standards	AI auto-evaluates signal quality and delivers direct conclusions

Reduce Labor Costs: Team members don't need deep network expertise — AI effectively closes the experience gap. New staff get up to speed quickly, significantly shortening training time.

Reduce Travel Costs: Most issues can be identified and resolved remotely. On-site visits are reserved for scenarios where physical presence is truly necessary.

Generate Additional Revenue (for MSPs / System Integrators): Inspection reports can be delivered directly as client service deliverables, forming a standardized, billable value-added service.

6. Transparent and Controllable — Every Step Visible

Every AI operation is fully transparent. Not a black box — a visible, accountable assistant.

Safety Mechanism	Description
✅ Read-only operations unrestricted	Query status, view alerts, analyze signal — zero risk, use anytime
✅ Write operations require confirmation	Modify config, upgrade firmware, reboot — changes displayed in full before execution, awaiting user confirmation
✅ Operations tied to user account	Every action is logged with the operator's identity; complete activity audit trail
✅ Standardized execution flow	Each operation follows the same process every time, regardless of who performs it — reducing human error risk

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7. Industry Applications

Applicable wherever distributed devices need centralized management.

Industry	Typical Scenario	Core Value
Retail Chain	Convenience stores, cafes, pharmacies	Network downtime directly impacts sales — fast recovery minimizes revenue loss
ATM / Financial	Bank branches, payment terminals	Strict compliance requirements and SLA commitments
Industrial / Energy	Production lines, power stations, mining sites	Remote O&M reduces downtime and ensures production continuity
Traffic Monitoring	Highways, urban traffic systems	Centralized batch management of hundreds or thousands of devices
MSP / System Integrator	Multi-customer managed services	Manage multiple clients with one person; inspection reports become billable services

Case Study: Retail Chain Store Network Operations *(Each store's router handles POS + surveillance + digital price tags + HQ connectivity)*

Scenario	Traditional Approach	With AI Network Assistant
Store outage — manager reports network down	Remote page-by-page investigation, time-consuming	AI locates root cause in minutes → **Significantly reduces business downtime**
Monthly inspection — days of device-by-device review	Manual device-by-device audit	AI generates complete report in minutes → **Problem stores automatically flagged**
Pre-event network check — ahead of major promotion	Manual monitoring, ad hoc	Real-time monitoring during event period → **O&M team has full visibility and confidence**
New store rollout — push standard config to all new devices	Manual device-by-device configuration	Natural language batch instruction → **Dramatically reduces labor investment**

8. License List

Module	Feature	InCloud Essential Plan	InCloud Branch Professional Plan
Device Management	Remote Configuration	√	√
	Firmware Upgrade	√	√
	Remote Reboot	√	√
	Remote Factory Reset	√	√
	Batch Management	√	√

Module	Feature	InCloud Essential Plan	InCloud Branch Professional Plan
Device Monitoring	Online/Offline	√	√
	Interface Status	√	√
	Link Monitoring	√	√
	Performance Status	√	√
	Cellular Signal	√	√
	Traffic Statistics	√	√
	Geolocation	√	√
	Alerts	√	√
Network & SD-WAN	Connector	×	√
	SD-WAN	×	√
Client Management	Connection Status & History	√	√
	Traffic Statistics	√	√
	POS Identification	√	√
	Out-of-Band Management	√	√
Remote O&M and Diagnostics	Remote Access	√	√
	Ping	√	√
	Traceroute	√	√
	Packet Capture	√	√
	Domain Monitoring	√	√
	Speed Test	√	√
	Log Download	√	√
AI-Powered O&M	Intelligent Inspection	√	√
	Fault Analysis	√	√
	Intelligent Report	√	√
Mobile	InCloud Manager App	√	√

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